

Similarity Among Variants: Output-Variant Correspondence

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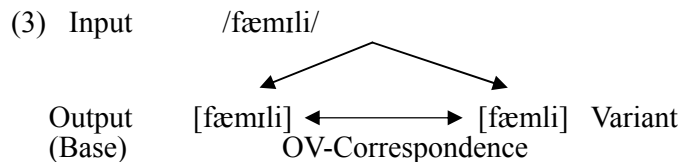
Main Goals

- a. To propose a principled account of opacity problems (Kiparsky 1973) involved in variant formations within the framework of Optimality Theory (Prince and Smolensky 1993). The existence of correspondence among variants is argued for, which is dubbed here ‘OV(Output-Variant)-Correspondence.’
- b. To present a model that generates variants without reranking or multiple strata, building upon Kurisu’s Realizational Morpheme Theory (2001).
- c. To critically assess the multi-stratal OT model.
- d. To compare OV-approach with local conjunction approach.

1. Preliminary.

(1) Variant and Base: When there is more than one output for one input (e.g., [fæmli] and [fæmli] for ‘family’), the one that is most faithful to the underlying representation is defined as “base”. Other output forms are collectively referred to as “variants”.

(2) OV-correspondence: A variant and a base are required to be identical by violable and ranked constraints. This is one kind of surface-surface correspondence (McCarthy 1995; Ito, Mester, Kitagawa 1996; Kenstowicz 1995, 1998; Ito and Mester 1997, 1998; Benua 1997 among others).



2. Case Studies

☞ Variants are often opaque. This is due to the requirement that a variant be phonologically identical to its base.

☞ The ranking Faith-OV (>>) Markedness >> Faith-IO gives rise to opacity. More concretely, (cf., McCarthy and Prince 1995, 1999; Benua 1997):

Underapplication: Faith-OV >> Markedness >> Faith-IO

Overapplication: Faith-OV, Markedness >> Faith-IO

2.1. Japanese

2.1.1 Japanese Postnasal Voicing (Ito and Mester 1995ab, 1999a; Ito, Mester and Padgett 1995 among others).

(4) Alternation due to *NT

Stem	Gerundive	Past	Gloss
a. tabe-	tabe-te	tabe-ta	‘eat’
b. hasir-	hasit-te	hasit-ta	‘run’
c. sin-	sin-de	sin-da	‘die’
d. kam-	kan-de	kan-da	‘bite’

(5) *NT: A sequence of a nasal and a voiceless obstruent is prohibited.

(6) Ident-IO-[voi]: Do not change the specification for [voi] in input-output mapping.

☞ Postnasal obstruents must be [+voiced] (in the native vocabulary).

☞ From the alternation of /t/ into [d], *NT >> Ident-IO-[voi]

(7) Violation of *NT in a variant formation

Base	Variant		Gloss
[anata]	[anta]	*anda	‘you’
[nanika]	[nan̩ka]	*nanga	‘something’
[nani+to]	[nanto]	*nando	‘with what’

☞ Variants produced by way of syncope systematically fail to obey *NT.

☞ This indicates Ident-IO-[voi] >> *NT. Hence we face a ranking paradox.

Proposal: What outranks *NT is not Ident-IO-[voi], but Ident-OV-[voi].

(8) OV-Correspondence between [anata] and [anta]

Base	[a	n	a	t	a]
Variant	[a	n		t	a]

(9)

Input /anata/ BASE [anata]	Ident-OV-[voi]	*NT	Ident-IO-[voi]
a. ☞ anta		*	
b. anda	*!		*

(10)

/sin+te/	Ident-OV-[voi]	*NT	Ident-IO-[voi]
a. sin+te	N/A	*!	
b. $\Rightarrow$ sin+de	N/A		*

Recursive Evaluation (Benua 1997)

$\Rightarrow$ A base form must be given in the form of output for the evaluation of a variant form, as seen in (9) (the voicing specification of post-nasal obstruents can not be given in input cf. underspecification, the Richness of the Base).

$\Rightarrow$ The overapplication in the base form (i.e., [anada] and [anda]) would be more harmonic than the underapplication in the variant form (i.e., [anata] and [anta]) as the former strategy avoids the violation of Ident-OV-[voi] as well as the markedness constraint, *NT.

(11) Recursion A

/anata/	Ident-OV-[voi]	*NT	Ident-IO-[voi]	>>
a. $\Rightarrow$ [anata]				
b. [anada]			*!	

Recursion B

>>		Ident-OV-[voi]	*NT	Ident-IO-[voi]
	a. $\Rightarrow$ [anta]		*	
	b. [anda]	*!		*

$\Rightarrow$ Since the evaluation of the base form takes precedence over the evaluation of the variant, the base form is unaffected by Faith-OV.

$\Rightarrow$ The base form is available in the form of output at the time of evaluation of the variant.

The emergence of the marked

(12)	Ranking Schema Faith-OV >> Markedness >> Faith-IO
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$\Rightarrow$ Marked structures (e.g., [nt]-sequence) are tolerated in the context of variant formations, but not in IO-dimensions (cf. McCarthy and Prince 1994).

2.1.2. Japanese *DD (Kuroda 1965; McCawley 1968; Ito and Mester 1995ab, 1999a; Lombardi 1998)

(13) *DD: No geminate of a voiced obstruent.

(14)	a. /uka(-uka)/	[ukkari]		absentmindedly
	/biku(-biku)/	[bikkuri]		surprised
	b. /zabu(-zabu)/	[zamburi]	*zabburi	raining heavily
	/nobi(-nobi)/	[nombiri]	*nobbiri	leisurely

☞ Intensification is expressed by way of internal gemination along with suffix /-ri/.

☞ As shown in (14b), gemination of a voiced obstruent is prohibited; instead, a nasal is inserted. Hence *DD >> Dep-IO-[nas]

(15) Violation of *DD in variant formations

Base	Variants		Gloss
[yabai]	[yabbai]	*[yappai]	‘dangerous’
	?*[yambai]		
[mazi(de)]	[mazzi(de)]	*[maʃʃi (de)]	‘seriously’
	*[manzi(de)]		
[huzakeru]	[huzzakeru]	*[hussakeru]	‘kid around’
		*[hunzakeru]	
[ʃoboi]	[ʃobboi]	*[ʃoppoi]	‘puny’
		??[ʃomboi]	

☞ In variant formation, a geminate of a voiced obstruent is allowed.

☞ Dep-OV-[nas] >> *DD >> Dep-IO-[nas].

(16)

Input /mazi/ BASE [mazi]	GEMINATE	Dep-OV-[nas]	*DD	Dep-IO-[nas]
a. ☞ mazzi			*	
b. manzi		*!		*
c. mazi	*!			

(17)

Input /zabu+ri/	GEMINATE	Dep-OV-[nas]	*DD	Dep-IO-[nas]
a. ☞ zamburi				*
b. zabburi			*!	
c. zaburi	*!			

2.2.1. Sea Dayak (Kenstowicz and Kisseberth 1979: 298; see also McCarthy 2002)

- ☞ Nasal harmony: nasality spreads to the following vowels up to a non-glide consonant.
- ☞ Postnasal Obstruent Deletion (Optional): in (d), [g] after [ŋ] can be optionally deleted.
- ☞ In a variant [nãŋa?], nasal harmony fails to apply (hence Ident-OV-[nas] >> *NV[oral])

- ☞ Approximant Coda Devoicing as in (19). Hence CODADEVOICE >> Ident-IO-[voi].
- ☞ Optional apocope as in (20).
- ☞ But a variant generated by apocope fails to undergo coda devoicing. Hence Ident-OV-[voi] >> CODADEVOICE

☞ In variant forms, an underlyingly pre-consonantal nasal must have a same place as its base

(i.e., it is not realized as the default form despite that is prevocalic).

☞ Nasal place assimilation overapplies in variant forms. Note that the trigger sound is not present at the surface level.

☞ This overapplication is accounted for by Ident-OV-[place] >> Ident-IO-[place], Place Markedness Sub-Hierarchy (*LAB, *DORS >> *COR; Prince and Smolensky 1993; Beckman 1998; Alderete et al 1999; Lombardi 1998, 2001 among many others), which determines the default place of articulation.

/N+gomo/ Base [ŋgomo]	Ident-OV-[place]	Ident-IO-[place]	*Dors	*Lab	*Cor
a. ☞ [ŋomo]		*	*		
b. [nomo]	*!	*			*
c. [momo]	*!			*	
d. [Nomo]	*!				

☞ Alternative approaches might be to utilize Max-IO-[place] (cf. Lombardi 1998, 2001) or to analyze this opacity with recourse to coalescence (cf. the analysis of Indonesian by Pater 1999).

2.2.4. Cairene Arabic Stress (McCarthy 1979)

☞ A superheavy ultima (CVCC, CVVC) is always stressed. A heavy ultima is not stressed. There is one systematic, though rare, exceptional pattern:

- (22)
- | | | | |
|-------------|----------------|---------------|---------------|
| a. katábt | ‘I write’ | sakakíin | ‘knives’ |
| b. sjarátun | ‘tree (nom)’ | sajaratuhúmaa | ‘their tree’ |
| c. ramáa | ‘he threw him’ | ʔaxúu | ‘his Brother’ |

☞ But according to McCarthy (1979: 452), all the forms “with a stressed final long vowel are in stylistically-conditioned variation with forms with final *h*: *ramáah*, *ʔaxúuh*.” This is analyzable as the effect of “prosodic faithfulness” (see the discussion in McCarthy 2000).

2.3 Emergence of the Marked in Syllable Structures

2.3.1. English

☞ There are various conditions on a complex onset in English (Kahn 1967; Selkirk 1982, 1984, Clements 1990 among others)

- (23) a. Two obstruents cannot constitute a complex onset except when the first member is /s/ (e.g., *stop*, *sky* but **ptight*, **tky*)
 b. only /s/ may appear with /m/ and /n/ (e.g., *smite*, *snare*, but **tmite*, **knight*)
 c. Two sonorants cannot form a complex onset (**mrite*, **lrik*).

☞ Optional syncope (Zwicky 1972; Hammond 1997, 1999 and many others), however, generates the structure that violates the condition above:

(24)		Regular form	Syncopated Form	Remarks
	potato	[pterto]	[p terto]	Violation of (23a)
	pachinko	[pətʃɪŋko]	[p tʃɪŋko]	Violation of (23a)
	connect	[konekt]	[k nekt]	Violation of (23b)
	tomato	[tomerto]	[t merito]	Violation of (23b)
	Marina	[mərɪnə]	[m rɪnə]	Violation of (23c)

☞ Assuming that the constraint that is responsible for the conditions in (23) is Sonority Sequencing Principle (SSP), Max-OV >> SSP >> Max-IO.

2.3.2. Ganda (Cole 1962)

☞ The language usually allows syllabic consonant only word-initially. The only exception is when optional apocope deletes the final vowel, in which case [m] can be syllabic:

(25)	Base	Variants	
	[ffumu]	[ffum̩]	‘spear’
	[mudumu]	[mudum̩]	‘pitcher’
	[ntamu]	[ntam̩]	‘clay cooking pot’

☞ Hence, Faith-OV >> *Syllbic Nasal >> Faith-IO (whatever the exact nature of involved Faith is)

2.3.3. Etsako (Elimelech 1976; Rosenthal 1994, 1997)

☞ The language generally prohibits a long vowel and a diphthong. It exhibits a long vowel and a diphthong only as a consequence of optional /l/-deletion:

(26)	[sele]	[see]	‘to climb’
	[lolo]	[loo]	‘to imitate’
	[okphali]	[okphai]	‘basket’
	[elumhi]	[eumhi]	‘copper’

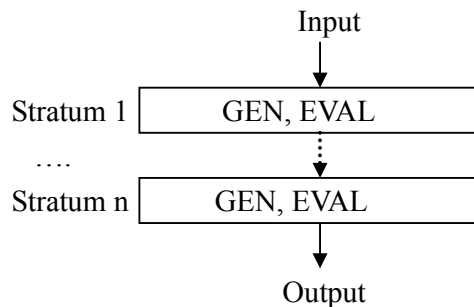
3. Theoretical Issues: Reranking and Multiple Strata

3.1. Lexical Phonology

3.1.1. Model

☞ An alternative to solve the problems presented here is Multi-Stratal OT (serial OT) (McCarthy and Prince 1993, Chapter 3; Inkelas and Orgun 1995; Sprouse 1997; Kiparsky 1998 and others).

(27) Multi-Stratal Model

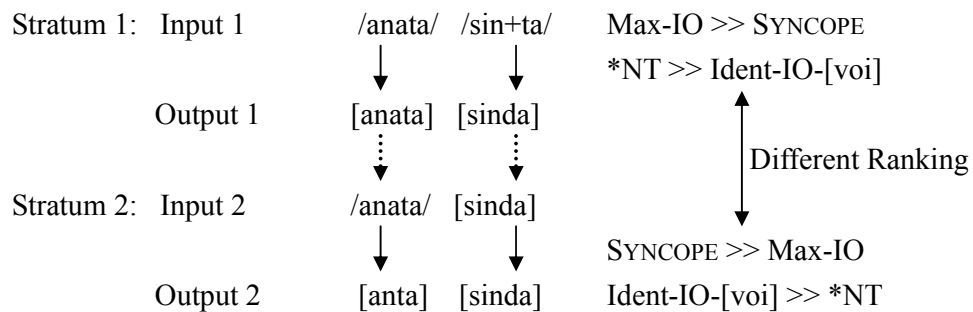


☞ A grammar is internally structured as serially ordered strata. Each stratum has its own ranking i.e., allows language-internal reranking.

☞ Here a two-level grammar suffices. Reranking at Stratum 2 is optional.

☞ A variant form is generated when reranking takes place at Stratum 2.

(28) Underapplication of *NT



☞ Syncope takes place at Stratum 2 (due to the reranking of SYNCOPE and Max).

☞ Ident-IO-[voi] is promoted over *NT at Stratum 2.

(29) Evaluation at Stratum 1 for /anata/

/anata/	Max-IO	SYNCOPE	*NT	Ident-IO-[voi]
a.  [anata]		*		
b. [anada]		*		*!
c. [anta]	*!		*	
d. [anda]	*!			*

(30) Evaluation at Stratum 1 for /sin+ta/

/sin+ta/	Max-IO	SYNCOPE	*NT	Ident-IO-[voi]
a.  [sinda]		*		*
b. [sinta]		*	*!	
c. [snta]	*!		*	
d. [snda]	*!			*

(31) Evaluation at Stratum 2 for /anata/

/anata/	SYNCOPE	Max-IO	Ident-IO-[voi]	*NT
a.  [anta]		*		*
b. [anda]		*	*!	
c. [anata]	*!			
d. [anada]	*!		*	

(32) Evaluation at Stratum 2 for /sinta/

/sinda/	*Complex	SYNCOPE	Max-IO	Ident-IO-[voi]	*NT
a. [sinta]		*		*!	*
b.  [sinda]		*			
c. [snta]	*!		*	*	*
d. [snda]	*!		*		

3.1.2. Against Multi-Stratal OT

 Serial OT treats opacity essentially as an accidental product of promotion of faithfulness constraints. OV-correspondence on the other hand captures opacity as the realization of demand that a variant be similar to its base (cf. function of opacity to preserve underlying phonological contrast).

 Serial OT faces many additional questions: why reranking is necessary in the first place?; how many strata are necessary?

 Serial OT predicts that the ranking of each stratum can be radically different. This seems empirically unsupported (Benua 1997).

 The two approaches drastically differ in their predictions. OV-approach only enhances similarity between a variant and its base, but never promotes disparity. On the other hand, the serial OT grammar can enhance the discrepancy between a variant and its base.

3.2. Variants without Reranking.

Two Crucial Observations

☞ a variant often has a sociolinguistic meaning (e.g., [sugoi] and [suge:], the latter being a bit condescending).

☞ a variant formation exhibits an “anti-faithful” effect (Alderete 1999, 2001; Horwood 1999; Kurisu 2001) and “anti-markedness” effects (e.g., gemination).

==> *Hypothesis*: A variant formation is morphologically driven.

3.2.1. Realizational Morphology Theory.

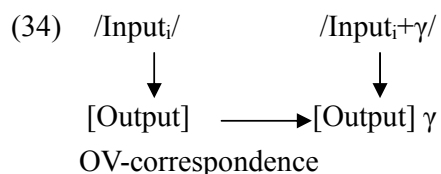
(33) Emergence of Non-additive Morphology (Kurisu 2001: 59)

Faith α >> RM >> Faith β

☞ RM (Realize Morpheme) requires every morpheme to be phonologically realized.

☞ RM >> Faith induces nonconcatenative morphology (Max for subtractive morphology, Ident for umlaut etc.) i.e., phonological deformation takes place for the satisfaction of RM.

3.2.2. Remodeling of OV-correspondence



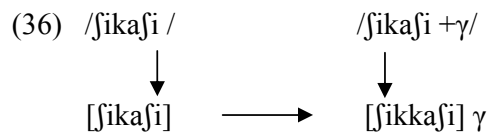
☞ A variant often has an additional sociolinguistic meaning (e.g., a rough flavor). Hence a variant is underlyingly ‘base+ sociolinguistic morpheme γ ’

☞ γ is required to be phonologically realized by RM and thus a variant has a different phonological shape from its base.

☞ OV-correspondence is now reduced to one specific kind of OO-correspondence (Benua 1997).

3.2.3 Example

(35)	Base	Variants	Gloss
	[baka]	[bakka]	‘stupid’
	[aho]	[ahho]	‘idiot’
	[ʃikaʃi]	[ʃikkaʃI]	‘but’



(37)

/ʃikaʃi +γ/	RM	Dep-μ- γ
a. [ʃikaʃi]	*!	
b. ☞ [ʃikkaʃi]		*

☞ RM >> Dep-μ- γ causes gemination to realize γ.

☞ Neither the promotion of markedness constraints or faithfulness constraints can account for gemination (unless one stipulates a constraint like GEMINATE).

☞ A variant often has a sociolinguistic meaning. Multi-stratal approach must stipulate that when reranking takes place at Stratum 2, a sociolinguistic meaning is added.

3.3. Against Partial Order Theory (most extensively advanced by Anttila 2000, 2002; Anttila and Cho 1998)

☞ A grammar includes partial order of constraints, and this is the source of variation.

☞ The criticisms to Multi-Stratal OT directly apply to this theory:

- anti-markedness effect cannot be derived by partial order theory.
- the addition of sociolinguistic meaning is inexplicable.

☞ This theory cannot distinguish markedness arising in IO-dimensions from markedness arising from OV-relations. Consider the English syncope example. The otherwise prohibited structures e.g., pt, kn, mr arise only from OV-relations.

=> Partial Order Theory predicts that this would also take place in IO-dimensions, contrary to fact.

4. Local Conjunction

4.1. What is local conjunction

☞ Another alternative account might be local conjunction (Smolensky 1993, 1995, 1997; Kirchner 1996; Lubowicz 2002; Ito and Mester 1998, 1999b among others).

☞ There are cases where despite that $C_1 \gg C_2$, C_3 is motivated independently, the simultaneous violation of C_2 and C_3 seems to be more important (i.e., ranked higher) than C_1 .

(38) Japanese: no more than one voiced obstruent within one prosodic word

- a. [futa], [buta] but no such words as [buda]
- b. [nise+tanuki] => [nisedanuki] (*Rendaku*)
- c. [takara+kuji] => [takarakuji] but not *[takaraguji] (Layman's Law)

☞ In Japanese (except foreign borrowings), more than one voiced obstruent is not allowed within one word (cf., Layman's Law (Ito and Mester 1986 etc)).

☞ Since one obstruent is allowed to surface, *VoiObs >> Ident-IO-[voi]. This, however, fails to account for the gap above.

☞ The proposal by Ito and Mester (1998): the self-conjunction of *VoiObs is ranked higher than Ident-IO-[voi]

(39) { *VoiObs & *VoiObs }_{wd} >> Ident-IO-[voi] >> *VoiObs

(40a)

/fuda/	{ *VoiObs & *VoiObs } _{wd}	Ident-IO-[voi]	*VoiObs
a. ☞ [fuda]			*
b. [futa]		*!	

(40b)

/buda/	{ *VoiObs & *VoiObs } _{wd}	Ident-IO-[voi]	*VoiObs
a. [buda]	*!		**
b. ☞ [buta]		*	*

4.2. Accounting for the opacity problem

- Isthmus Nahuatl

- (41) a. ʃikakíli or ʃikakíl *ʃikakíl̥ 'put it in it'
- b. kítaja or kítaj *kítaj̥ 'he already sees it'
- c. kikówa or kików *kików̥ 'a season'

☞ Variants already incur the violation of Max because of the apocope.

☞ Though CODADEVoice >> Ident-IO-[voi] is motivated because coda devoicing does take effect, it might be the case that the conjunction of Max and Ident-IO-[voi] is ranked higher than CODADEVoice:

(42)

/ʃikakili/	{Max-IO & Ident-IO-[high]}	DEVoice	Ident-IO-[voi]	Max-IO
a. ☞ [ʃikakíl]		*		*
b. [ʃikakíl̥]	*!		*	*

4.3. Comparison

☞ This approach, however, cannot account for (i) overapplication and (ii) “partial” underapplication.

☞ In the case of overapplication, the reference to levels other than input-output dimension is necessary (i.e., the conjunction of any IO-faithfulness cannot account for the problem). Hence local conjunction that exclusively refers to IO-dimension cannot account for such opaque phenomena. See the overapplication of Nasal Place Assimilation in Mwera.

☞ Local conjunction approach makes a wrong prediction concerning “partial” underapplication. Consider the data from Sea Dayak:

- (43) a. mǎta ‘an eye’
b. mǎã ‘the face’
c. mǎjǎ ‘a season’
- d. /naŋga/ [naŋga?] or [nǎŋa?] ‘straighten’
*[nǎŋǎ?]

(44)

/naŋga?/	{Ident-IO-[nas] & Max-IO}	*NV[oral]	Ident-IO-[nas]	Max-IO
a. 𐀀 [naŋa?]		**		*
b. [nǎŋǎ?]	*!		**	*
c. (☞) [nǎŋa?]	*!	*	*	*

☞ The local conjunction predicts that the phonological process is blocked altogether.

☞ In short, OV-Faith approach has a wider empirical coverage.

5. Summary

- Opacity is often involved in variant formations. This strongly motivates the existence of OV-correspondence.
- Intermediate levels can be obviated in favor of OV-correspondence.
- RMT can be applied to variant formations. Based upon this, OV-correspondence is reduced to one type of OO-correspondence.
- Though local conjunction system can account for some of the problems, OV-faith approach covers wider range of phenomena concerning opacity in variants.

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